

RxJS Observable Composition

A compact reference for choosing higher-order mapping operators in Angular.

Start with behavior

Choose an operator by asking what should happen when the source emits again before the current inner Observable completes. This makes the decision easier to explain and test.

switchMap

Cancels the previous inner subscription when a new value arrives. It is a strong default for type-ahead search and route-driven loading where stale responses should not win.

concatMap

Queues inner Observables and preserves their order. Use it when each operation must complete before the next one begins.

exhaustMap

Ignores new source values while an inner Observable is active. This is useful for preventing duplicate login, payment or form submissions.

mergeMap

Allows inner Observables to run concurrently. Use it for independent work, while considering concurrency limits and error behavior.

Cleanup and errors

Keep error handling close to the operation that can recover. Long-lived streams should have an explicit lifecycle strategy, such as `takeUntilDestroyed` in Angular.

Operator	When a new value arrives	Typical use
switchMap	Cancel previous	Search and route loading
concatMap	Queue it	Ordered saves
exhaustMap	Ignore it	Duplicate-submit prevention
mergeMap	Run concurrently	Independent requests